

Where Does GPT-2 Keep Subject–Verb Agreement When Heads Are Pruned?

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Abstract

This report studies subject–verb agreement in GPT-2 Medium under head pruning. From 1000 requested BLiMP examples, 995 valid subject–verb aligned sentences remain after Stanza dependency filtering and GPT-2 token alignment. We run four linked analyses: (RQ1) expert head resilience, (RQ2) linear decodability from residual updates (syntactic ghosting), (RQ3) low-dimensional functional taxonomy, and (RQ4) behavioral stability with transition-style comparisons. Across all analyses, agreement information remains recoverable under strong pruning, supporting a distributed residual-stream account rather than a strictly head-local account.

1 Motivation for the research problem

Subject–verb agreement is a clean syntax test. To predict the correct verb form, the model must preserve number information from the subject across distance and intervening words. This makes agreement useful for testing where syntactic information is stored in transformer internals.

Two broad explanations are possible. In a head-local account, a small set of attention heads carries most syntactic routing, so pruning these heads should sharply reduce agreement signal. In a distributed account, many components contribute partial information and the residual stream keeps a robust aggregate representation, so agreement can survive major pruning.

This distinction matters in two ways. Theoretically, it clarifies how to interpret mechanistic claims about syntax in language models. Practically, it informs whether pruning-based compression can be safely applied to grammar-sensitive tasks.

Research problem. Where is subject–verb agreement represented in GPT-2 Medium under progressive head pruning, and does number information remain linearly decodable from residual updates Δh when most heads are removed?

Objectives. (O1) Identify and track expert heads (verb→subject attention) and test whether agreement signal persists in Δh across pruning levels.
(O2) Characterize global head behavior through low-dimensional feature structure and sentence-level stability analysis.

2 Hypotheses and predictions

- H1.** A small subset of heads has high verb→subject attention mass and remains behaviorally stable while active.
- H2.** Number remains linearly decodable from Δh above chance even with strong pruning, while participation ratio $\rho(\Delta h)$ decreases.
- H3.** Head features lie in a low-dimensional space and form interpretable clusters.
- H4.** Expert behavior is stable under bootstrap resampling, and sentence difficulty (especially length) affects syntax-hit rates.

How hypotheses are tested. H1 is tested with expert trajectories across pruning milestones (RQ1). H2 is tested with probe accuracy and ρ trends across milestones (RQ2). H3 is tested with PCA and clustering over head-level features (RQ3). H4 is tested with bootstrap confidence intervals and sentence-stratified comparisons (RQ4).

3 Methods

Data. We use BLiMP English subject–verb agreement subsets. From 1000 requested examples, 995 valid aligned sentences remain after dependency filtering and token alignment. The same aligned set is used across all analyses to keep RQ comparisons consistent.

Model. GPT-2 Medium (24 layers, 16 heads/layer, hidden size 1024, head size 64), sequence length 64, attentions and hidden states enabled.

Head importance and pruning (RQ1–RQ2). Head importance is measured by single-head ablation: increase in causal LM loss when one head output is zeroed via a forward hook. Baseline loss on the aligned set is 6.835607. Heads are ranked by this score and pruned to active-head milestones $M \in \{300, 250, 100, 50\}$ out of 384.

```

attn_output = output[0]                                     # (B, S, D)
reshaped = attn_output.view(B, S, num_heads, head_dim)
reshaped = reshaped * layer_mask.view(1, 1, num_heads, 1)
attn_output = reshaped.view(B, S, D)

```

Listing 1: Core pruning operation used in `pruner_model.py`.

RQ1: Expert tracking. Experts are top heads by verb→subject mass with one-head-per-layer diversity. At each milestone we record entropy and syntax-match (verb-row argmax on subject token).

RQ2: Residual ghosting. For layer ℓ , $\Delta h_\ell = h_\ell - h_{\ell-1}$ at the last non-padding token. Participation ratio is

$$\rho(\Delta h) = \frac{(\sum_k \lambda_k)^2}{\sum_k \lambda_k^2}, \quad (1)$$

where λ_k are squared singular values of centered updates.

Linear probe with 80/20 stratified split:

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N \left[y_i \log \sigma(\mathbf{w}^\top \Delta h_i + b) + (1 - y_i) \log(1 - \sigma(\mathbf{w}^\top \Delta h_i + b)) \right]. \quad (2)$$

RQ3: Functional taxonomy. Each head is represented by 18 standardized features (distance, entropy, delimiter/prev/next/self mass, offset bins, syntax-match), then analyzed with PCA and k -means ($k = 4$).

RQ4: Stability and transitions. For each expert and sentence we compute syntax-hit and dependency mass. Bootstrap with 1000 resamples gives 95% CIs. We also compare short vs. long and simple vs. complex sentence groups, and run transition-style rule matching.

Inference method. H2 is supported if probe accuracy stays high at low M while ρ declines. H4 is supported by stable bootstrap intervals plus interpretable sentence-group differences.

4 Results

4.1 RQ1: Expert heads under pruning

Top experts on 995 aligned sentences are: L2H9 (0.805), L8H6 (0.507), L11H6 (0.506), L10H12 (0.485), L13H3 (0.393), L7H6 (0.361), L21H14 (0.318), L14H7 (0.313), L15H7 (0.304), and L20H15 (0.286). These ten heads come from ten distinct layers, showing distributed specialization.

At $M = 300$ and $M = 250$, heads such as L8H6 and L10H12 stay active and increase syntax-match. At $M = 100$, several high baseline experts are already pruned, but strong tracked behavior remains. At $M = 50$, most experts are pruned, yet high syntax-match traces still appear for selected heads.

Table 1: Representative expert syntax-match under pruning ($n = 995$).

Head	Baseline	$M = 300$	$M = 250$	$M = 100$	$M = 50$
L2H9	0.805 (ACT)	0.833 (PRU)	0.833 (PRU)	0.822 (PRU)	0.825 (PRU)
L8H6	0.507 (ACT)	0.581 (ACT)	0.592 (ACT)	0.614 (ACT)	0.501 (PRU)
L11H6	0.506 (ACT)	0.469 (ACT)	0.470 (ACT)	0.491 (PRU)	0.472 (PRU)
L13H3	0.393 (ACT)	0.407 (ACT)	0.418 (ACT)	0.437 (PRU)	0.392 (PRU)

4.2 RQ2: Syntactic ghosting in residual updates

Layer 3 is the strongest decodability band throughout pruning. At baseline ($M = 384$), layer-3 accuracy is 96.98% with PR 25.04. At $M = 250$, layer-3 accuracy remains 95.48% with PR 8.68. At $M = 100$, it remains 92.96% with PR 5.40. At $M = 50$, it remains 91.46% with PR 2.82.

So dimensionality shrinks strongly while linear decodability stays high, consistent with H2.

[EN] RQ1 — Functional Resilience Under Pruning

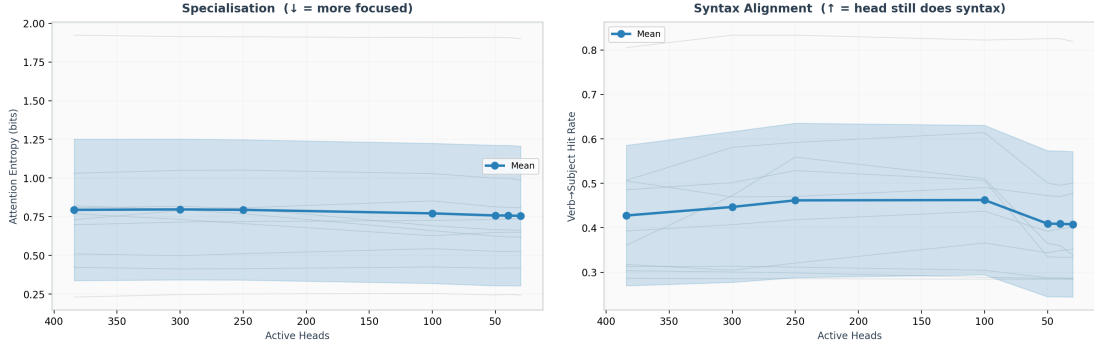


Figure 1: RQ1 expert entropy and syntax-match trajectories.

Table 2: Layer-3 participation ratio and probe accuracy across pruning ($n = 995$).

Active heads M	$\rho(\Delta h_3)$	Probe acc. (layer 3)
384	25.04	96.98%
300	10.47	93.97%
250	8.68	95.48%
100	5.40	92.96%
50	2.82	91.46%

[EN] RQ2 — Syntactic Ghosting in the Residual Stream

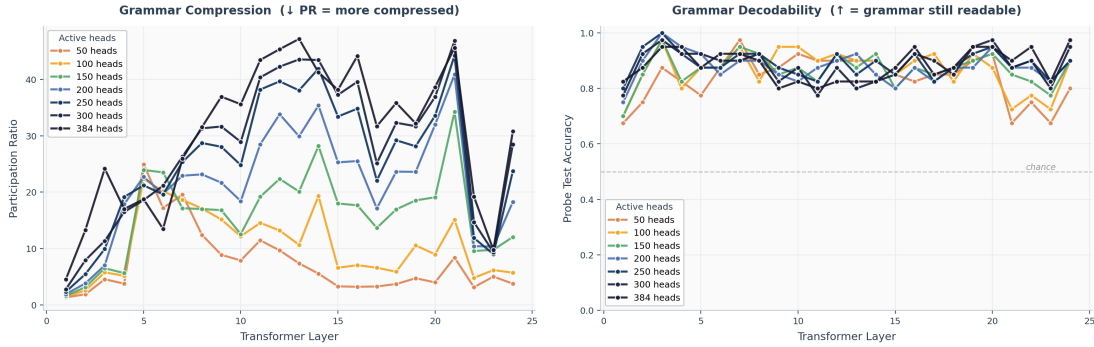


Figure 2: RQ2 participation-ratio and probe trends under pruning.

4.3 RQ3: Functional head taxonomy

PCA shows clear low-dimensional structure: PC1 explains 46.6%, PC2 explains 22.8%, and 4 PCs reach 90% cumulative variance. For $k = 4$, cluster counts are Syntactic 248, Local/Positional 59, Global/Diffuse 57, and Delimiter 20. Local and Delimiter clusters are concentrated in early layers; Syntactic and Global clusters appear deeper.

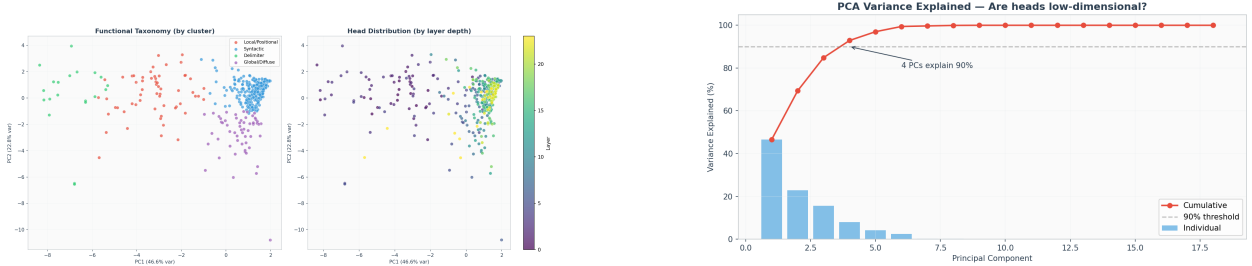


Figure 3: RQ3 clustering geometry (left) and PCA variance profile (right).

4.4 RQ4: Stability and transition analysis

Bootstrap analysis identifies L2H9 as strongest expert: mean 0.805, 95% CI [0.780, 0.829]. Other high heads are L8H6 (0.507), L11H6 (0.506), and L10H12 (0.485).

Length effect is clear: short syntax-hit 0.433 vs. long 0.348. Complexity split in this run is flat/slightly reversed: simple 0.427 vs. complex 0.432.

Transition-rule assignment is concentrated: previous-token 100.0%; all other synthetic rules 0.0%.

Table 3: Representative bootstrap means and CIs ($B = 1000$, $n = 995$).

Head	Mean	95% CI	Var.
L2H9	0.805	[0.780, 0.829]	0.1570
L8H6	0.507	[0.476, 0.536]	0.2500
L11H6	0.506	[0.476, 0.537]	0.2500
L10H12	0.485	[0.451, 0.516]	0.2498

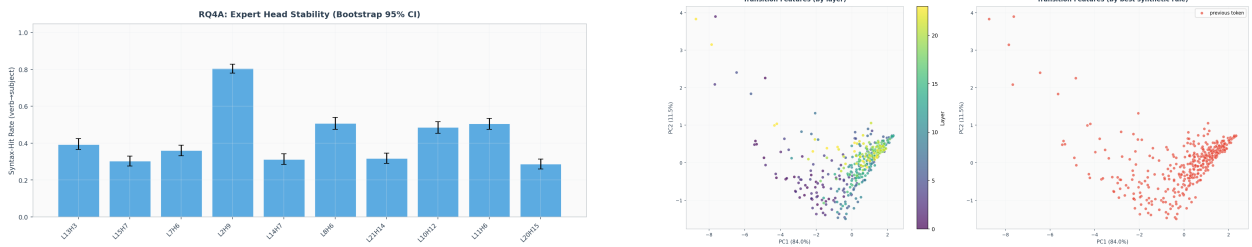


Figure 4: RQ4 bootstrap stability (left) and transition-feature analysis (right).

5 Theoretical implications

Implications for language processing theory. The short-vs-long gap is directionally consistent with dependency-distance and retrieval-interference ideas. The complexity split is less consistent in this run, which suggests that complexity effects depend on how complexity is defined and sampled.

Implications for machine processing. Agreement remains linearly recoverable from residual updates under heavy pruning. This supports a distributed representation view and suggests that pruning evaluation should include residual-level probes, not only attention visualizations.

Implications for interpretability practice. RQ1 and RQ2 together argue against a strictly head-local explanation. RQ3 shows recurring functional motifs across heads. RQ4 shows that global transition signatures can be strongly positional even when specific heads have high syntax-match.

Limitations and future verification. This report studies one architecture family and one English agreement slice. A direct extension is to rerun the same RQ1–RQ4 template across model sizes and alternative architectures to test generality of residual robustness. Additional languages are useful when the research question shifts from architecture behavior to typological comparison.

Table 4: Future verification roadmap for architecture behavior.

Target rerun	What it verifies
GPT-2 small / large	Scale sensitivity of expert concentration, ρ compression, and residual decodability.
Other decoder-only LMs	Whether residual ghosting under pruning generalizes beyond GPT-2.
Alternative pruning scores	Robustness of findings to importance metric choice.
Larger/diverse agreement sets	Stability of rankings and cluster proportions under broader coverage.

References

- Warstadt, A., et al. (2020). BLiMP: The Benchmark of Linguistic Minimal Pairs for English. *TACL*, 8, 377–392.
- Voita, E., et al. (2019). Analyzing Multi-Head Self-Attention: Specialized Heads Do the Heavy Lifting, the Rest Can Be Pruned. *ACL*, 5797–5808.
- Elhage, N., et al. (2021). A Mathematical Framework for Transformer Circuits. *Transformer Circuits Thread*.
- Clark, K., et al. (2019). What Does BERT Look at? *BlackboxNLP*, 276–286.
- Gibson, E. (2000). The dependency locality theory. *Image, Language, Brain*, 95–126.
- Goldberg, Y. (2019). Assessing BERT’s Syntactic Abilities. *arXiv:1901.07931*.

Appendix A: Code and reproducibility

GitHub repository

<https://github.com/VaibhavKalyan/CGS-410>

Python scripts

These files implement RQ1–RQ4:

- `data_loader.py` — BLiMP load, Stanza parsing, GPT-2 token alignment
- `pruner_model.py` — ablation scoring and pruning hooks

- `rq1_functional_resilience.py` — RQ1 expert analysis
- `rq2_syntactic_ghosting.py`, `ghosting_probes.py` — RQ2 residual/probe analysis
- `rq3_functional_taxonomy.py` — RQ3 PCA and clustering
- `rq4_behavioral_stability.py`, `attention_metrics.py` — RQ4 stability and transitions

Example commands

- `python rq1_functional_resilience.py -lang en -max-sentences 1000`
- `python rq2_syntactic_ghosting.py -lang en -max-sentences 1000`
- `python rq3_functional_taxonomy.py -lang en -max-sentences 1000`
- `python rq4_behavioral_stability.py -lang en -max-sentences 1000`